aquaponics

AQUAPONICS controller: build manual

> **Status: control and forecasting core host-tested (13 checks); nothing built or run on live fish.** No new PCB: it runs on the BioTray board (`biotray/`, order-ready). ALL BIOLOGICAL LIMITS ARE PLACEHOLDERS: tune them with an aquaculture expert.

1. Parts

BioTray board (see `biotray/MANUFACTURING_MANUAL.md`); DS18B20 waterproof probe; analog pH module + probe; analog EC module + probe; analog dissolved-oxygen module + probe; ADS1115 breakout on the I2C header; 12 V circulation pump; 12 V dosing (peristaltic) pump + pH-down reservoir; relay module for the aerator (the FAN output drives the relay coil, flyback is on the board); backup battery aerator (strongly recommended); GFCI-protected supply; sealed enclosure for the wet environment.

2. Build

1. Build and test the BioTray board first. 2. Wire PUMP1 = circulation, PUMP2 = dosing, FAN = aerator relay coil, sensors as in `aquaponics/docs/DESIGN.md`. 3. Compile `aquaponics/firmware/src/main_aquaponics.cpp` with `firmware/lib/aq_core.c` (Arduino-ESP32 2.x): **the firmware compiles for the ESP32-S3 (tools/esp32_build.sh, 0 warnings, binary in `aquaponics/firmware/release/`); bring it up on a bench.** 4. Run `cd aquaponics/firmware && make test` on a PC (13 checks).

3. Calibrate

pH with pH 4 and 7 buffers, EC with a standard solution, DO with its 0 % and saturated points, then set the scale factors (placeholders in the skeleton). Set band limits for your species.

4. Commissioning

Dry run with water only and no fish for at least a week: dosing never exceeds the daily cap, a pulled probe stops dosing (bad-reading lockout), aeration follows the DO forecast, power-cut behaviour is safe. See `aquaponics/docs/TEST_PLAN.md`.

5. Limits

No ammonia/nitrite/nitrate sensing: test with a kit daily at first. Straight-line forecasts cannot foresee pump failure or feeding spikes. Mains aerators need a proper relay and a rated enclosure. Not tested on live animals. General rules: `manuals/STANDARD_PROCEDURES.md`.